

HR_Motor API Description

Return Value Format

- The first return value is the error code.
- The second return value is the error message.
- If the command is a read command with return values, the values are provided in `result[2...]`. Currently, only configuration and event retrieval commands return values.

Error Codes

The error code is placed in the first position, and the error message is placed in the second position.

```
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enum Code {
    kOk = 0,
    kExecuteFailed = 60001, // Server execution failed
    kInvalidCommand = 60002, // Invalid command
    kInvalidArgument = 60003, // Invalid argument
    kInvalidConfig = 60004, // Invalid configuration, including file read failure
    kMotorIsMoving = 60005, // Motor is running
    // Internal controller error codes
    kHRApplNotExist = 20800, // HRAppl does not exist
    kHRApplTimeOut = 20801, // HRAppl timeout
    // Others are unknown errors
};
```

If the return is not one of those listed, it should be displayed as an unknown error.

Global Management

Command	Parameters	Returns	Remarks
GetMotorList	motor_list: string[]	Gets all device names	

Example in Python:

```
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HRApCmd(0, "HR_Motor", "GetMotorList", [], result)
# result[0]: normally "0"
# result[1]: normally "OK"
# result[2...]: list of Motor names
```

Motor Control

Format: Command + Device Name (e.g., "J7") + Parameters + Return Values

Example in Python:

```
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HRApCmd(0, "HR_Motor", "MotorConnect", ["J7"], result)
HRApCmd(0, "HR_Motor", "MotorMovePosition", ["J7", "1.0"], result)
```

Connection Management

Command	Parameters	Returns	Remarks
MotorConnect	motor: string	Connects	
MotorDisconnect	motor: string	Disconnects	

State Management

Command	Parameters	Returns	Remarks
MotorGetState	motor: string	state: int	Get motor state: -1 = UNKNOWN, 0 = STOPPED, 1 = RUNNING

Command Execution

Command	Parameters	Returns	Remarks
MotorMovePositive	motor: string	Moves in positive direction	
MotorMoveNegative	motor: string	Moves in negative direction	
MotorMovePosition	motor: string, position: double	Moves to absolute position	
MotorMoveOrigin	motor: string	Finds origin	
MotorStop	motor: string	Stops all movements	
MotorRobotSafePosition	motor: string, speed: double	Returns robot to a safe position using a temporary speed, without modifying the overall configuration. Suitable for use in a programming environment.	

Command with Temporary Speed:

Command	Parameters	Returns	Remarks
MotorMovePositionSpeed	motor: string, position: double, speed: double	Moves to absolute position with specified speed	

Error Clearing Functionality:

Command	Parameters	Returns	Remarks
MotorCleanError	motor: string	Clears errors	